

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8	(a)			A is permanent hard water (1) it is not softened by boiling (only by ion exchange) / boiling has no effect on the volume of soap needed (1) B contains both temporary and permanent hard water (1) as it is partly softened by boiling <u>and</u> further softened by ion exchange / less soap needed after boiling and less again after ion exchange (1)	2		2	4		4
	(b)			$\text{Na}_2\text{CO}_3 + \text{MgCl}_2 \rightarrow 2\text{NaCl} + \text{MgCO}_3$ award (1) for reactants award (1) for products award (1) for balancing - can only be awarded if reactants and products are correct accept multiples of correct balancing ignore state symbols		3		3		
				Question 8 total	2	3	2	7	0	4